

Proposed title: *A Comparative and Explainable Deep-Learning Framework for Brain Tumor Detection, Classification, and Segmentation in MRI*

Research gaps and expected contribution

The clearest gap is not another single-model accuracy claim. It is the lack of standardized, clinically oriented comparison across models, datasets, tasks, and validation settings. Reviews repeatedly identify inconsistent validation, insufficiently diverse data, overfitting, limited interpretability, and weak clinical usability. A paper that uses patient-level splits, independent testing, transparent preprocessing, calibrated probabilities, and localization-aware explanations would address these weaknesses directly.

A second gap is the separation between benchmark classification and clinical decision support. Several studies report very high accuracy on public datasets—for example, 98.7% for a CNN comparison across two MRI datasets and 99.33% for a fine-tuned EfficientNet-B4 study that also used cross-validation and an independent blind test.

Such results justify further investigation, but they do not establish clinical readiness unless the model is evaluated on representative external data, compared with qualified readers, tested prospectively or retrospectively in realistic workflows, and assessed for calibration and failure behaviour.

A third gap is the limited treatment of efficiency and privacy. Lightweight architectures may be useful where computation is constrained, while federated learning may help institutions collaborate without directly pooling patient data; both directions are identified in the segmentation literature as avenues for addressing deployment and privacy challenges.

So our paper can therefore make a stronger contribution by reporting a performance–efficiency–interpretability trade-off rather than optimizing one headline metric.